The documentation and process conversion measures necessary to comply with this revision shall be completed by 8 December 2021.

INCH-POUND

MIL-PRF-19500/614L
w/AMENDMENT 1
8 September 2021
SUPERSEDING
MIL-PRF-19500/614L
30 August 2018

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, FIELD EFFECT RADIATION HARDENED
ENCAPSULATED (THROUGH-HOLE AND SURFACE MOUNT PACKAGES)
N-CHANNEL, SILICON, TYPES 2N7380, AND 2N7381,
JANTXV AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for an N-channel, radiation hardened (total dose and single event effects (SEE)), enhancement mode, MOSFET, power transistors intended for use in high density power switching applications. Two levels of product assurance (JANTXV and JANS) are provided for each device type as specified in MIL-PRF-19500, with avalanche energy ratings ($E_{AS}$) and maximum avalanche current ($I_{AS}$). See 6.7 for JANHC and JANKC die versions.

1.2 Package outlines. The device package outlines are as follows: TO-257AA in accordance with figure 1 and SMD-0.5 TO-276AA (U3) in accordance with figure 2 for all encapsulated device types. The dimensions and topography for JANHC and JANKC unencapsulated die are as listed in specification sheet MIL-PRF-19500/657.

1.3 Maximum ratings. Unless otherwise specified, $T_C$ = +25°C.

| Type | $P_T$ (1) $T_C$ = +25°C | $P_T$ $T_A$ = +25°C (free air) | $R_{\theta JC}$ (2) | Min $V_{(BR)DSS}$ $V_{GS}$ = 0 V $I_D$ = 1.0 mA dc | $I_{D1}$ (3) (4) $T_C$ = +25°C | $I_{D2}$ (3) (4) $T_C$ = +100°C | $T_J$ and $T_{STG}$ |
|---|---|---|---|---|---|---|---|
| | W | W | °C/W | V dc | A dc | A dc | °C |
| 2N7380, 2N7380U3 | 75 | 2 | 1.67 | 100 | 14.4 | 9.1 | -55 to +150 |
| 2N7381 | 75 | 2 | 1.67 | 200 | 9.4 | 6.0 | -55 to +150 |

| Type | $I_S$ | $I_{DM}$ (5) | $V_{GS}$ | $E_{AS}$ max | $I_{AS}$ |
|---|---|---|---|---|---|
| | A dc | A(pk) | V dc | mJ | A dc |
| 2N7380, 2N7380U3 | 14.4 | 57.6 | ±20 | 150 | 14.4 |
| 2N7381 | 9.4 | 37.6 | ±20 | 150 | 9.4 |

(See notes next page)

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A
FSC 5961

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

1.3 Maximum ratings. continued

(1) Derate linearly by 0.6 W/°C for $T_C$ > +25°C.
(2) See figure 3, thermal impedance curves.
(3) The following formula derives the maximum theoretical $I_D$ limit. $I_D$ is limited by package and device construction.

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

(4) See figure 4, maximum drain current graph.
(5) $I_{DM}$ = 4 X $I_{D1}$ as calculated in note (3).

1.4 Primary electrical characteristics. Unless otherwise specified, $T_C$ = +25°C.

| Type | Min $V_{(BR)DSS}$ $V_{GS}$ = 0 $I_D$ = 1.0 mA dc | $V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ $I_D$ = 1.0 mA dc | | $I_{DSS}$ max $V_{GS}$ = 0 | Max $r_{DS(on)1}$ (1) $V_{GS}$ = 12 V; $I_D$ = $I_{D2}$ | |
|---|---|---|---|---|---|---|
| | | | | $V_{DS}$ = 80 percent of rated $V_{DS}$ | $T_J$ = +25°C | $T_J$ = +150°C |
| | V dc | V dc | | µA dc | Ω | Ω |
| | | Min | Max | | | |
| 2N7380, 2N7380U3 | 100 | 2.0 | 4.0 | 25 | 0.18 | 0.36 |
| 2N7381 | 200 | 2.0 | 4.0 | 25 | 0.40 | 0.84 |

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.5 for PIN construction example and 6.6 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JANTXV” and "JANS".

1.5.2 JAN brand and quality level designators for unencapsulated devices (die). See 6.7 for unencapsulated devices.

* 1.5.3 Radiation hardness assurance (RHA) designator. The RHA levels that are applicable for this specification sheet from lowest to highest are as follows: "M", "D", "R", "F".

1.5.4 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.4.1 First number and first letter symbols. The transistors of this specification sheet are identified by the first number and letter symbols "2N".

1.5.4.2 Second number symbols. The second number symbols for the transistor covered by this specification sheet are as follows: "7380” and “7381”.

1.5.4.3 Suffix letters. No suffix letters are used on devices that are packaged in the TO-257AA package of figure 1. The suffix letters "U3" are used on devices that are packaged in the SMD-0.5 TO-276AA package of figure 2.

1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

1.5.6 Unencapsulated devices. See 6.7 for JANHC and JANKC die versions.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

* 1.6 Radiation features:

Maximum total ionizing dose (TID) available (Dose rate = 50-300 rad(Si)/s)):

For device type 2N7380, 2N7380U3, 2N7381: ..... 500 krads(Si) 1/

Heavy Ion Single Event Effect (SEE) SEB and SEGR test:

For device type 2N7380, 2N7380U3:
No SEB and SEGR were observed at surface LET (see table IV) ..... ≤ 37 MeV·cm$^2$/mg 2/
(In-situ Bias $V_{DS}$ = 100 V and $V_{GS}$ = 0 V
(In-situ Bias $V_{DS}$ = 90 V and $V_{GS}$ = -5 V
(In-situ Bias $V_{DS}$ = 70 V and $V_{GS}$ = -10 V
(In-situ Bias $V_{DS}$ = 50 V and $V_{GS}$ = -15)

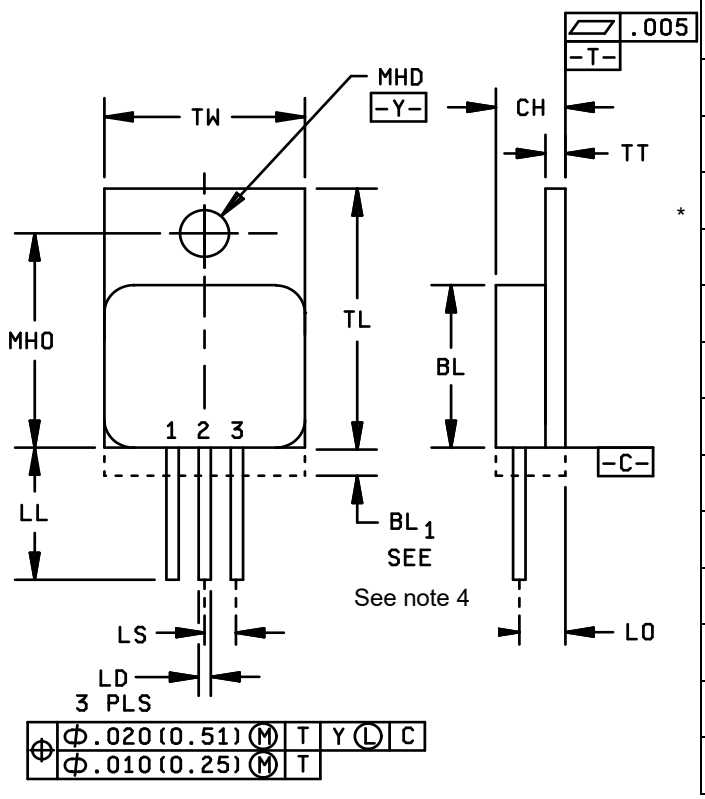
For device type 2N7381:
No SEB and SEGR were observed at surface LET (see table IV) ..... ≤ 37 MeV·cm$^2$/mg 2/
(In-situ Bias $V_{DS}$ = 100 V and $V_{GS}$ = -10 V
(In-situ Bias $V_{DS}$ = 50 V and $V_{GS}$ = -15 V)

1/ Manufacturer supplying device types 2N7380, 2N7380U3, and 2N7381 has performed characterization testing in accordance with MIL-STD-750, method 1019, condition A (dose rate = 50 - 300 rad(Si)/s). The radiation end point limits are guaranteed only for the conditions as specified in MIL-STD-750, method 1019, condition A to a maximum total ionizing dose level of 500 krads(Si).

2/ Manufacturer also performed heavy ion SEB and SEGR test at Brookhaven National Lab Accelerator for the MOSFET technology devices in accordance with TM1080 of MIL-STD-750. Limits are characterized at initial qualification and after any design or process changes which may affect the SEE (SEB/SEGR) characteristics. For more information on SEE (SEB/SEGR) test results, customers are requested to contact the manufacturer.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.



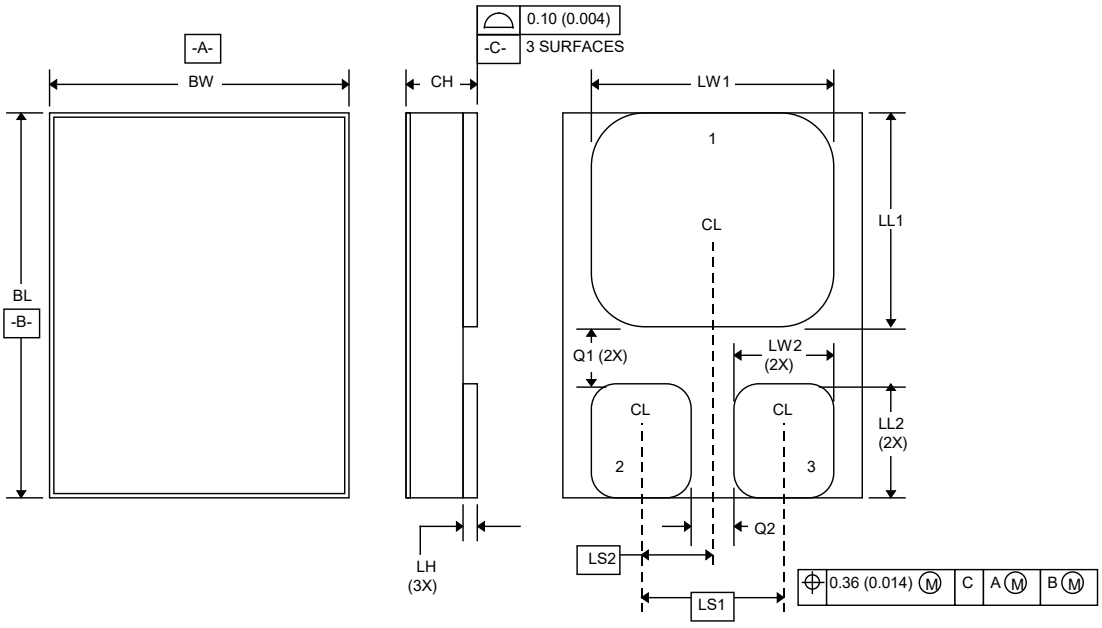

| Ltr | Inches | | Millimeters | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| BL | .410 | .430 | 10.41 | 10.92 |
| $BL_1$ | | .028 | | 0.71 |
| CH | .190 | .200 | 4.83 | 5.08 |
| LD | .025 | .035 | 0.64 | 0.89 |
| * LL | .500 | .650 | 12.70 | 16.51 |
| LO | .120 BSC | | 3.05 BSC | |
| LS | .100 BSC | | 2.54 BSC | |
| MHD | .140 | .150 | 3.56 | 3.81 |
| MHO | .527 | .537 | 13.39 | 13.64 |
| TL | .645 | .665 | 16.38 | 16.89 |
| TT | .035 | .045 | 0.89 | 1.14 |
| TW | .410 | .420 | 10.41 | 10.67 |
| Term 1 | Drain | | | |
| Term 2 | Source | | | |
| Term 3 | Gate | | | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. All terminals are isolated from case.
4. This area is for the lead feed-through eyelets (configuration is optional, but will not extend beyond this zone).
5. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.

* FIGURE 1. Dimensions and configuration (TO-257AA).



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.




| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .395 | .405 | 10.04 | 10.28 |
| BW | .291 | .301 | 7.40 | 7.64 |
| CH | | .124 | | 3.15 |
| LH | .010 | .020 | 0.25 | 0.51 |
| LW1 | .281 | .291 | 7.14 | 7.39 |
| LW2 | .090 | .100 | 2.29 | 2.54 |
| LL1 | .220 | .230 | 5.59 | 5.84 |
| LL2 | .115 | .125 | 2.93 | 3.17 |
| LS1 | .150 BSC | | 3.81 BSC | |
| LS2 | .075 BSC | | 1.91 BSC | |
| Q1 | .030 | | 0.762 | |
| Q2 | .030 | | 0.762 | |
| TERM 1 | Drain | | | |
| TERM 2 | Gate | | | |
| TERM 3 | Source | | | |

NOTES:
1. Dimension are in inches.
2. Millimeters are given for information only.
3. The lid shall be electrically isolated from the drain, gate, and source.
* 4. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.

* FIGURE 2. Dimensions and configuration (TO-276AA, SMD-0.5).



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.
* MIL-STD-883 - Test Methods Standard Microcircuits.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

$I_{AS}$ ..... Rated avalanche current, non-repetitive.
nC ..... nano Coulomb.
PIND ..... Particle impact noise detector.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (TO-257AA) and figure 2 (TO-276AA) herein. Methods used for electrical isolation of the terminal feedthroughs shall employ materials that contain a minimum of 90 percent $Al_2O_3$ (ceramic). Examples of such construction techniques are metallized ceramic eyelets or ceramic walled packages.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction shall not be permitted to meet the requirements of this specification.

* 3.4.3 Silicone Die coating. The use of a silicone die coat requires a successful completion of MIL-STD-883, method 5011 on each silicone lot for its intended applications, and as part of the full MIL-PRF-19500 qualification process.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrostatic discharge sensitive (ESDS). The devices covered by this specification sheet have been classified as ESDS. The devices shall be handled in accordance with the ESD program established to comply with the requirements of MIL-PRF-19500 to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source. $R \leq 100\ k\Omega$, whenever bias voltage is to be applied drain to source.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.8 Electrical test requirements. The electrical test requirements shall be table I as specified herein.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.2.1.1 Single event effects (SEE). SEE (SEB/SEGR) shall be performed in accordance with TM1080 of MIL-STD-750 at initial qualification and after process or design changes which may affect radiation hardness (see table III and table IV). Upon qualification, manufacturers shall provide the verification test conditions from section 5 of method 1080 of MIL-STD-750 that were used to qualify the device for inclusion into section 6 of the slash sheet. End-point measurements shall be in accordance with table II. SEE (SEB/SEGR) characterization data shall be made available upon request of the qualifying or acquiring activity.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

4.3 Screening (JANS and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table E-IV of MIL-PRF-19500) (1) (2) | Measurement | |
|---|---|---|
| | JANS level | JANTXV level |
| (3) | Gate stress test (see 4.3.1) | Gate stress test (see 4.3.1) |
| (3) | Method 3470 of MIL-STD-750. (see 4.3.2) | Method 3470 of MIL-STD-750. (see 4.3.2) |
| (3) 3c | Method 3161 of MIL-STD-750 (see 4.3.3) | Method 3161 of MIL-STD-750 (see 4.3.3) |
| 5 | Method 2052 of MIL-STD-750, PIND (see MIL-PRF-19500 and 4.3.4) | Not applicable |
| 9 | Subgroup 2 of table I herein | Not applicable |
| 10 | Method 1042 of MIL-STD-750, test condition B | Method 1042 of MIL-STD-750, test condition B |
| 11 | Subgroup 2 of table I herein.<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±25 µA dc or ±100 percent of initial value, whichever is greater. | Subgroup 2 of table I herein. |
| 12 | Method 1042 of MIL-STD-750, test condition A | Method 1042 of MIL-STD-750, test condition A |
| 13 | Subgroup 2 and 3 of table I herein.<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±25 µA dc or ±100 percent of initial value, whichever is greater.<br><br>$\Delta r_{DS(on)1}$ = ±20 percent of initial value.<br>$\Delta V_{GS(th)1}$ = ±20 percent of initial value. | Subgroup 2 of table I herein.<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±25 µA dc or ±100 percent of initial value, whichever is greater.<br><br>$\Delta r_{DS(on)1}$ = ±20 percent of initial value.<br>$\Delta V_{GS(th)1}$ = ±20 percent of initial value. |
| 17 | For TO-257AA packages: Method 1081 of (see 4.3.5), Endpoints: Subgroup 2 of table I herein | For TO-257AA packages: Method 1081 of (see 4.3.5), Endpoints: Subgroup 2 of table I herein |

(1) At the end of the test program, $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ are measured.
(2) An out-of-family program to characterize $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, and $V_{GS(th)1}$ shall be invoked.
(3) Shall be performed anytime after temperature cycling, screen 3a; JANTXV level does not need to be repeated in screening requirements.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

4.3.1 Gate stress test. Apply $V_{GS}$ = ±30 V minimum for t = 250 μs minimum.

4.3.2 Single pulse avalanche energy ($E_{AS}$).

a. Peak current ($I_{AS}$).....$I_{D1}$.

b. Peak gate voltage ($V_{GS}$).....12 V.

c. Gate to source resistor ($R_{GS}$).....$25 \leq R_{GS} \leq 200\ \Omega$.

d. Initial case temperature .....+25°C +10°C, -5°C.

e. Inductance ..... $\left[\frac{2E_{AS}}{(I_{D1})^2}\right]\left[\frac{V_{BR} - V_{DD}}{V_{BR}}\right]$ mH minimum.

f. Number of pulses to be applied .....1 pulse minimum.

g. Supply voltage .....$V_{DD}$ = 50 V, or 25 V for 100 V devices.

4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{MD}$, $t_{SW}$, (and $V_H$ where appropriate). See table III, group E, subgroup 4 herein.

4.3.4 PIND. Not applicable in screening when devices are processed using alternative method and flow requirements approved by the qualifying activity, that includes incorporating the use of certified clean processing and silicone die coat. Instead, the PIND test performance shall be performed in group B3 and group C3, on a lot sample basis. PIND failures detected in group B or C will represent lot jeopardy and be evaluated for root cause and lot integrity.

4.3.5 Dielectric withstanding voltage.

a. Magnitude of test voltage.....800 V dc for TO-257AA, 600 V dc for TO-276AA.

b. Duration of application of test voltage.....15 seconds (min).

c. Points of application of test voltage .....All leads to case (bunch connection).

d. Method of connection .....Mechanical.

e. Kilovolt-ampere rating of high voltage source .....1,200 V/1.0 mA (min).

f. Maximum leakage current.....1.0 mA.

g. Voltage ramp up time.....500 V/second

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JANTXV) of MIL-PRF-19500, and as follows.

4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1051 | Condition G. |
| B3 | 2052 | PIND, required if not performed in screening. (22 devices, c = 0 for large lots, 12 devices, c = 0 for small lots). |
| B4 | 1042 | The heating cycle shall be 1 minute minimum. |
| B5 | 1042 | Condition A; $V_{DS}$ = 100 percent of rated; $T_A$ = +175°C, t = 120 hours or $T_A$ = +150°C, t = 240 hours; read and record $V_{BR(DSS)}$ (pre and post) at $I_D$ = 1 mA; read and record $I_{DSS}$ (pre and post), in accordance with table I, subgroup 2. |
| B5 | 1042 | Condition B; $V_{GS}$ = 100 percent of rated $T_A$ = +175°C, t = 24 or $T_A$ = +150°C, t = 48 hours. |
| B6 | 3161 | $R_{\Theta JC}$(max) = 1.67°C/W (not applicable for U3 device) |

4.4.2.2 Quality levels JAN, JANTX and JANTXV, table E-VIB of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B2 | 1051 | Condition G. |
| B3 | 1042 | The heating cycle shall be 1 minute minimum. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition A, weight = 10 lbs (4.54 kg), t = 10 seconds (TO-257AA package only). |
| C3 | 2052 | PIND, required if not performed in screening. (JANS only, 22 devices, c = 0 for large lots, 12 devices, c = 0 for small lots). |
| C5 | 3161 | See 4.3.3, $R_{\theta JC(max)}$ = 1.67°C/W (not required if test was performed in group B) |
| C6 | 1042 | The heating cycle shall be 1 minute minimum. No heat sink nor forced air cooling on the device shall be permitted. |

4.4.4 Group D Inspection. Group D inspection shall be conducted in accordance with table E-VIII of MIL-PRF-19500 and table II herein.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical inspection | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance 2/ | 3161 | See 4.3.3 | $Z_{\theta JC}$ | | | °C/W |
| Breakdown voltage drain to source | 3407 | $V_{GS}$ = 0, $I_D$ = 1 mA dc, bias condition C | $V_{(BR)DSS}$ | | | |
| 2N7380, 2N7380U3 | | | | 100 | | V dc |
| 2N7381 | | | | 200 | | V dc |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1.0 mA | $V_{GS(th)1}$ | 2.0 | 4.0 | V dc |
| Gate current | 3411 | $V_{GS}$ = +20 V dc, $V_{DS}$ = 0, bias condition C | $I_{GSSF1}$ | | +100 | nA dc |
| Gate current | 3411 | $V_{GS}$ = -20 V dc, $V_{DS}$ = 0, bias condition C | $I_{GSSR1}$ | | -100 | nA dc |
| Drain current | 3413 | $V_{GS}$ = 0, $V_{DS}$ = 80 percent of rated $V_{DS}$, bias condition C | $I_{DSS1}$ | | 25 | µA dc |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = 12 V dc, condition A, pulsed (see 4.5.1), $I_D$ = rated $I_{D2}$ (see 1.3) | $r_{DS(on)1}$ | | | |
| 2N7380, 2N7380U3 | | | | | 0.18 | Ω |
| 2N7381 | | | | | 0.40 | Ω |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = 12 V dc, condition A, pulsed (see 4.5.1), $I_D$ = rated $I_{D1}$ (see 1.3) | $r_{DS(on)2}$ | | | |
| 2N7380, 2N7380U3 | | | | | 0.20 | Ω |
| 2N7381 | | | | | 0.49 | Ω |
| Forward voltage (source drain diode) | 4011 | Condition A, $V_{GS}$ = 0, $I_D$ = rated $I_{D1}$ | $V_{SD}$ | | | |
| 2N7380 | | | | | 1.8 | V dc |
| 2N7380U3 | | | | | 1.5 | V dc |
| 2N7381 | | | | | 1.4 | V dc |

See footnotes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +125°C | | | | |
| Gate current | 3411 | Bias condition C, $V_{GS}$ = ±20 V dc, $V_{DS}$ = 0 | $I_{GSS2}$ | | ±200 | nA dc |
| Drain current | 3413 | Bias condition C, $V_{GS}$ = 0, $V_{DS}$ = 80 percent of rated $V_{DS}$ | $I_{DSS2}$ | | 0.25 | mA dc |
| Static drain to source on-state | 3421 | Condition A, $V_{GS}$ = 12 V dc, pulsed (see 4.5.1), $I_D$ = rated $I_{D2}$ | $r_{DS(on)3}$ | | | |
| 2N7380, 2N7380U3 | | | | | 0.36 | Ω |
| 2N7381 | | | | | 0.75 | Ω |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1.0 mA dc | $V_{GS(th)2}$ | 1.0 | | V dc |
| Low temperature operation: | | $T_A$ = -55°C | | | | |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1.0 mA dc | $V_{GS(th)3}$ | | 5.0 | V dc |
| Subgroup 4 | | | | | | |
| Switching time test | 3472 | $I_D$ = rated $I_{D1}$, $V_{GS}$ = 12 V dc, gate drive impedance = 7.5 Ω, $V_{DD}$ = 50 percent of $V_{BR(DSS)}$ | | | | |
| Turn-on delay time | | | $t_{d(on)}$ | | | |
| 2N7380, 2N7381 | | | | | 25 | ns |
| 2N7380U3 | | | | | 35 | ns |
| Rise time | | | $t_r$ | | | |
| 2N7380 | | | | | 60 | ns |
| 2N7380U3 | | | | | 75 | ns |
| 2N7381 | | | | | 50 | ns |
| Turn-off delay time | | | $t_{d(off)}$ | | | |
| 2N7380, 2N7380U3 | | | | | 40 | ns |
| 2N7381 | | | | | 70 | ns |
| Fall time | | | $t_f$ | | | |
| 2N7380, 2N7380U3 | | | | | 30 | ns |
| 2N7381 | | | | | 60 | ns |

See footnotes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 4 - Continued. | | | | | | |
| Forward transconductance | 3475 | $I_D = I_{D2}$, $V_{DD}$ = 15 V dc see 4.5.1 | $g_{fs}$ | 2.5 | | s |
| Subgroup 5 | | | | | | |
| Safe operating area test (high voltage) | 3474 | See figure 5, $t_p$ = 10 ms, $V_{DS}$ = 80 percent of rated $V_{BR(DSS)}$, $V_{DS} \leq$ 200 V maximum | | | | |
| Electrical measurements | | See table I, subgroup 2 | | | | |
| Subgroup 6 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 7 | | | | | | |
| Gate charge | 3471 | Condition B | | | | |
| On-state gate charge | | | $Q_{g(on)}$ | | | |
| 2N7380, 2N7380U3 | | | | | 40 | nC |
| 2N7381 | | | | | 50 | nC |
| Gate to source charge | | | $Q_{gs}$ | | 10 | nC |
| Gate to drain charge | | | $Q_{gd}$ | | | |
| 2N7380, 2N7380U3 | | | | | 20 | nC |
| 2N7381 | | | | | 25 | nC |
| Reverse recovery time | 3473 | $di/dt \leq$ 100 A/µs, $V_{DD} \leq$ 50 V, $I_D = I_{D1}$ | $t_{rr}$ | | | |
| 2N7380, 2N7380U3 | | | | | 275 | ns |
| 2N7381 | | | | | 460 | ns |

1/ For sampling plan, see MIL-PRF-19500.
2/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 3 and 4 (JANS).
Group B, subgroups 2 and 3 (JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

TABLE II. Group D inspection.

| | Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Pre-irradiation limits | | | | Post-irradiation limits | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | M, D, P, L, and R | | F, G, and H 4/ | | M, D, P, L, and R | | F, G, and H 4/ | | |
| | | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| | Subgroup 2 | | $T_C$ = +25°C | | | | | | | | | | |
| * | Steady-state total dose irradiation ($V_{GS}$ bias) | 1019 | Condition A, $V_{GS}$ = 12 V, $V_{DS}$ = 0 | | | | | | | | | | |
| * | Steady-state total dose irradiation ($V_{DS}$ bias) | 1019 | Condition A, $V_{GS}$ = 0, $V_{DS}$ = 80 percent of rated $V_{DS}$ (pre-irradiation) | | | | | | | | | | |
| | End-point electrical: | | | | | | | | | | | | |
| | Breakdown voltage, drain to source | 3407 | $V_{GS}$ = 0, $I_D$ = 1 mA, bias condition C | $V_{(BR)DSS}$ | | | | | | | | | |
| | 2N7380, | | | | 100 | | 100 | | 100 | | 100 | | V dc |
| | 2N7380U3 | | | | 100 | | 100 | | 100 | | 100 | | V dc |
| | 2N7381 | | | | 200 | | 200 | | 200 | | 200 | | V dc |
| | Gate to source voltage (threshold) 4/ | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1 mA | $V_{GS(th)}$ | 2.0 | 4.0 | 2.0 | 4.0 | 2.0 | 4.0 | 1.25 | 4.5 | V dc |
| | Gate current | 3411 | $V_{GS}$ = 20 V, $V_{DS}$ = 0 V, bias condition C | $I_{GSSF1}$ | | 100 | | 100 | | 100 | | 100 | nA dc |
| | Gate current | 3411 | $V_{GS}$ = -20 V, $V_{DS}$ = 0, bias condition C | $I_{GSSR1}$ | | -100 | | -100 | | -100 | | -100 | nA dc |

See footnotes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

TABLE II. Group D inspection - Continued.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Pre-irradiation limits | | | | Post-irradiation limits | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Method | Conditions | | M, D, P, L, and R | | F, G, and H 4/ | | M, D, P, L, and R | | F, G, and H 4/ | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| Subgroup 2 - Continued | | $T_C$ = +25°C | | | | | | | | | | |
| Drain current | 3413 | $V_{GS}$ = 0, bias condition C, $V_{DS}$ = 80 percent of rated $V_{DS}$ (pre-irradiation) | $I_{DSS}$ | | | | | | | | | |
| 2N7380, 2N7380U3 | | | | | 25 | | 25 | | 25 | | 50 | µA dc |
| 2N7381 | | | | | 25 | | 25 | | 25 | | 50 | µA dc |
| Static drain to source on-state voltage | 3405 | $V_{GS}$ = 12 V, condition A pulsed, see 4.5.1. $I_D$ = $I_{D2}$ | $V_{DS(ON)}$ | | | | | | | | | |
| 2N7380, 2N7380U3 | | | | | 1.638 | | 1.638 | | 1.638 | | 2.184 | V dc |
| 2N7381 | | | | | 2.4 | | 2.4 | | 2.4 | | 3.18 | V dc |
| Forward voltage source drain diode | 4011 | $V_{GS}$ = 0, $I_D$ = $I_{D1}$, bias condition A | $V_{SD}$ | | | | | | | | | |
| 2N7380, | | | | | 1.8 | | 1.8 | | 1.8 | | 1.8 | V |
| * 2N7380U3 | | | | | 1.5 | | 1.5 | | 1.5 | | 1.5 | V |
| 2N7381 | | | | | 1.4 | | 1.4 | | 1.4 | | 1.4 | V |

1/ For sampling plan, see MIL-PRF-19500.
2/ Separate samples shall be pulled for each bias.
3/ Group D qualification may be performed anytime prior to lot formation. Wafers qualified to these group D QCI requirements may be used for any other specification sheet utilizing the same die design.
4/ The “H” designation represents devices which pass end-points at M, D, P, L, R, F, G, and H designated total-ionizing-dose (TID).



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

TABLE III. Group E inspection (all quality levels) - for qualification or re-qualification only.

| Inspection | MIL-STD-750 | | Sample plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices, c = 0 |
| Temperature cycling | 1051 | Test condition G, 500 cycles | |
| Hermetic seal<br>Fine leak<br>Gross leak | 1071 | | |
| Electrical measurements | | See table I, subgroup 2 | |
| Subgroup 2 1/ | | | 45 devices, c = 0 |
| Steady-state reverse bias | 1042 | Condition A, 1,000 hours | |
| Electrical measurements | | See table I, subgroup 2 | |
| Steady-state gate bias | 1042 | Condition B, 1,000 hours | |
| Electrical measurements | | See table I, subgroup 2 | |
| Subgroup 4 | | | Sample size N/A |
| Thermal impedance curves | | See MIL-PRF-19500. | |
| Subgroup 10 | | | 22 devices, c = 0 |
| Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors | 3476 | Test conditions shall be derived by the manufacturer | |
| Subgroup 11 | | | 3 devices |
| SEE 2/ 3/ | 1080 | See method 1080 of MIL-STD-750. | |

1/ A separate sample for each test shall be pulled.
2/ Group E qualification of SEE testing may be performed prior to lot formation. Qualification may be extended to other specification sheets utilizing the same structurally identical die design.
3/ Device qualification to a higher level LET is sufficient to qualify all lower level LETs.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

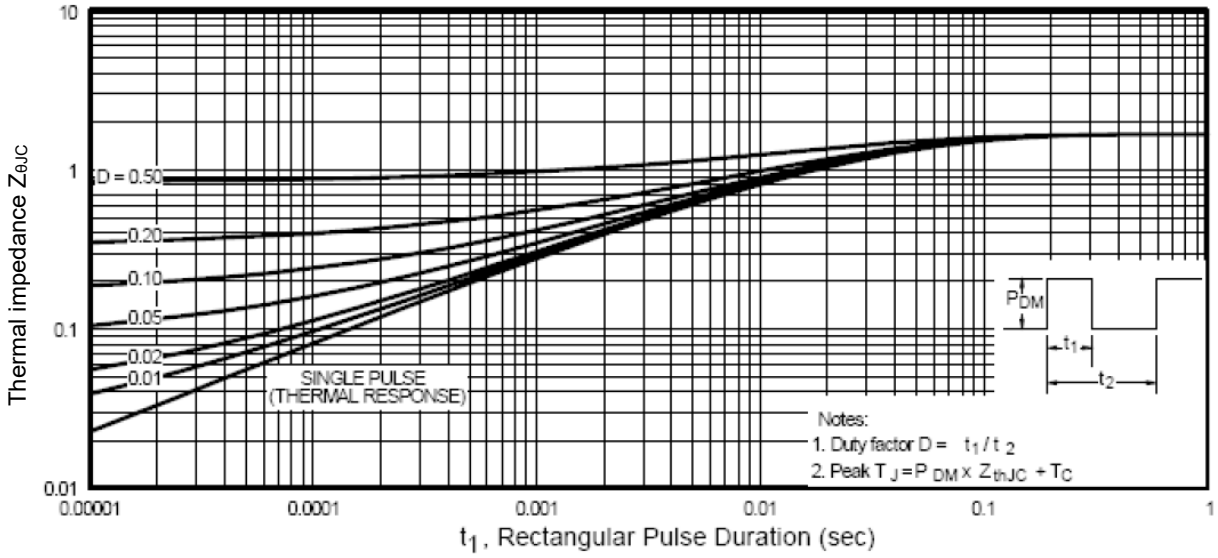
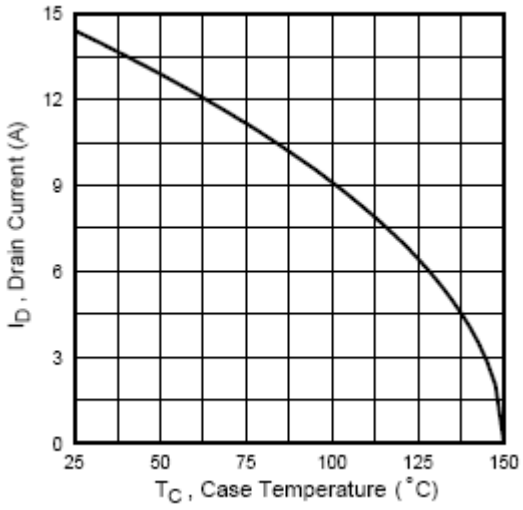



FIGURE 3. Thermal impedance curves.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.




2N7381

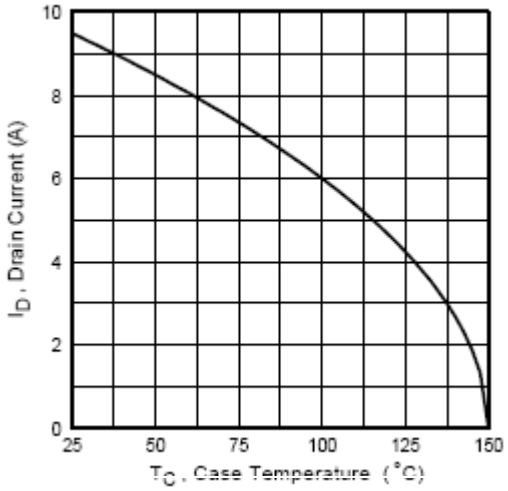
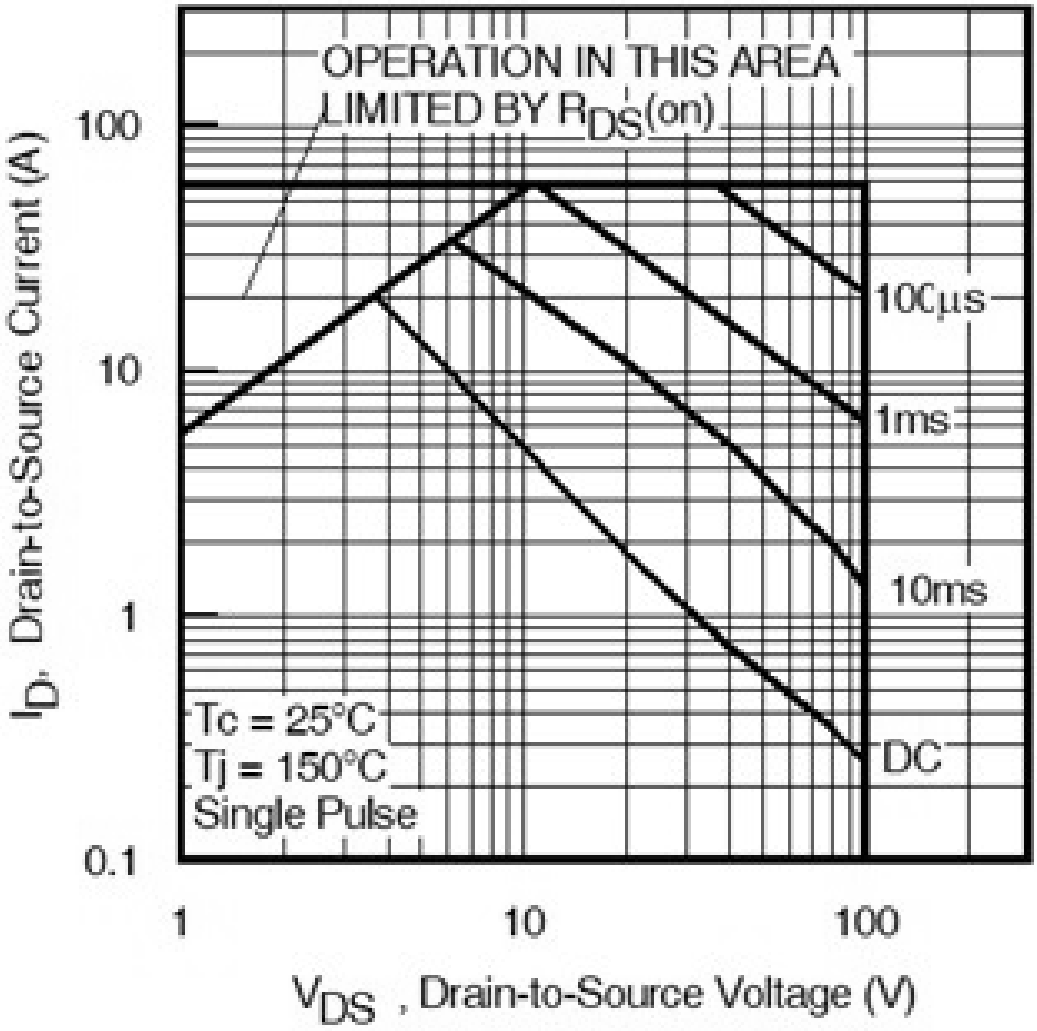



FIGURE 4. Derating drain current graphs.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

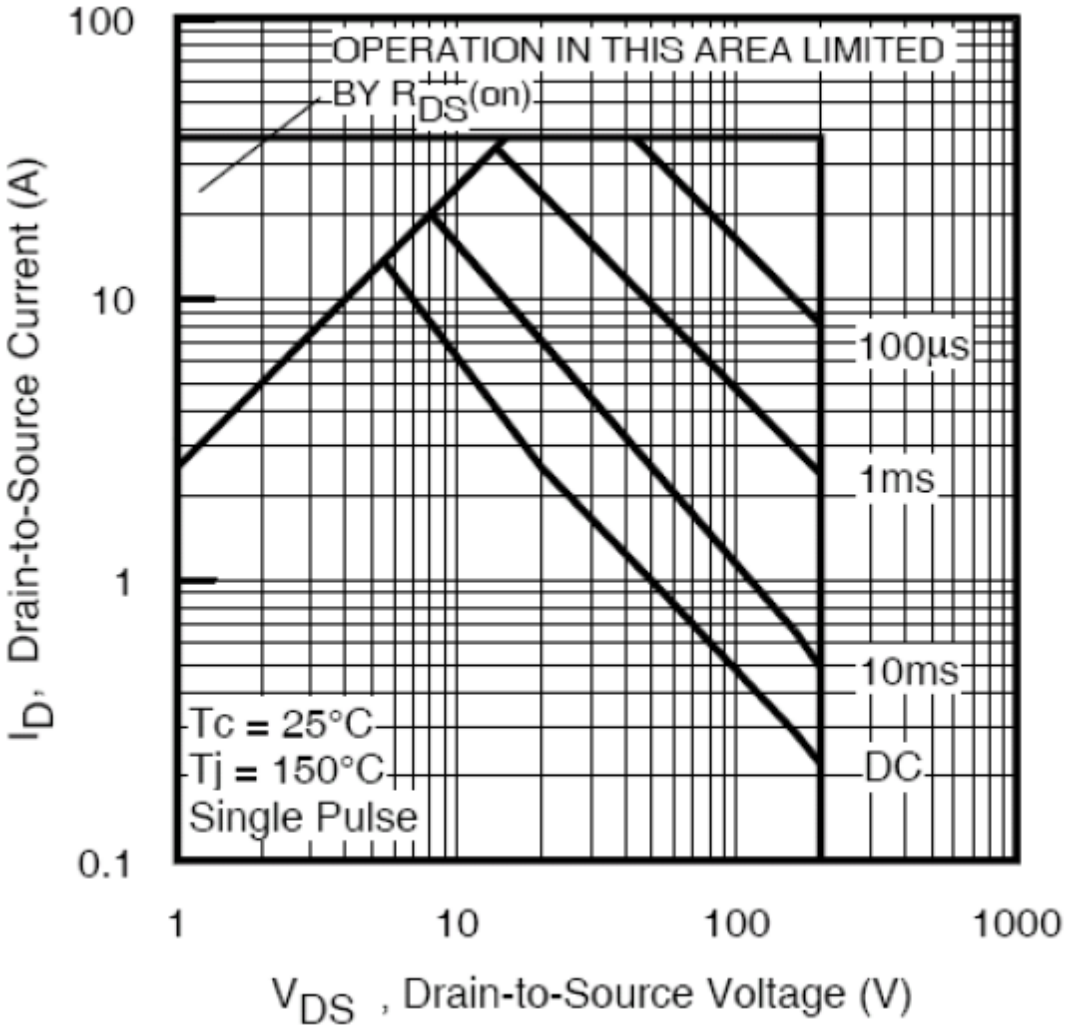



2N7380, 2N7380U3

* FIGURE 5. Safe operating area graphs.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.




2N7381

FIGURE 5. Safe operating area graphs - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete Part or Identifying Number (PIN), see 1.5 and 6.5.

e. For acquisition of RHA designated devices, table II, subgroup 1 testing of group D herein is optional. If subgroup 1 is desired, it should be specified in the contract.

f. If SEE testing data is desired, it should be specified in the contract or order.

g. If specific SEE characterization conditions are desired (see section 6.8 and table IV), manufacturer's CAGE code should be specified in the contract or order.

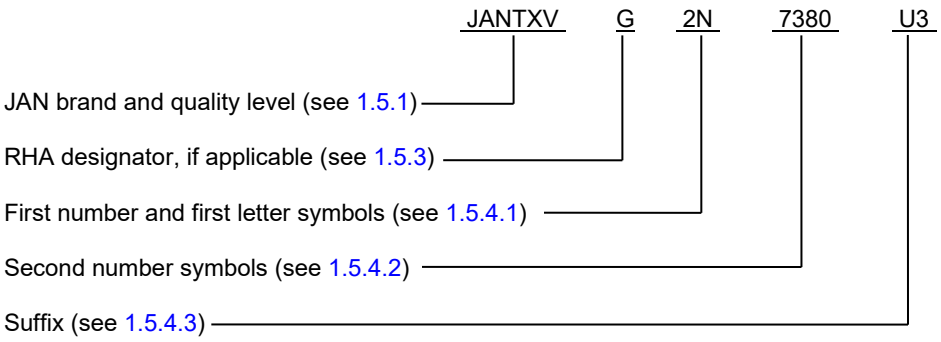
6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 Supersession data. This specification supersedes DESC drawing 89009, dated 19 December 1989.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

6.5 PIN construction example. The PINs for encapsulated devices are construction using the following form.




6.6 List of PINs. The following is a list of possible PINs available on this specification sheet.

| PINs for devices of the "TXV" quality level | PINs for devices of the "TXV" quality level with RHA (1) | PINs for devices of the "S" quality level | PINs for devices of the "S" quality level with RHA (1) |
|---|---|---|---|
| JANTXV2N7380 | JANTXV#2N7380 | JANS2N7380 | JANS#2N7380 |
| JANTXV2N7380U3 | JANTXV#2N7380U3 | JANS2N7380U3 | JANS#2N7380U3 |
| JANTXV2N7381 | JANTXV#2N7381 | JANS2N7381 | JANS#2N7381 |

(1) The number sign (#) represent one of eight RHA designators available (M, D, P, L, R, F, G, or H). The PIN is also available without a RHA designator.

6.7 JANC die versions. The JANHC and JANKC die versions of these devices are covered under specification sheet MIL-PRF-19500/657.

6.8 Application data.

* 6.8.1 Manufacturer specific irradiation data. Each manufacturer qualified to this slash sheet has characterized its devices to the requirements of MIL-STD-750 method 1080 and as specified herein. Since each manufacturer's characterization conditions can be different and can vary by the version of method 1080 qualified to, the MIL-STD-750 method 1080 revision version date and conditions used by each manufacturer for characterization have been listed here (see table IV) for information only. SEE (SEB and SEGR) conditions and figures listed in section 6 are current as of the date of this specification sheet, please contact the manufacturer for the most recent conditions.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

TABLE IV. Manufacturers characterization conditions.

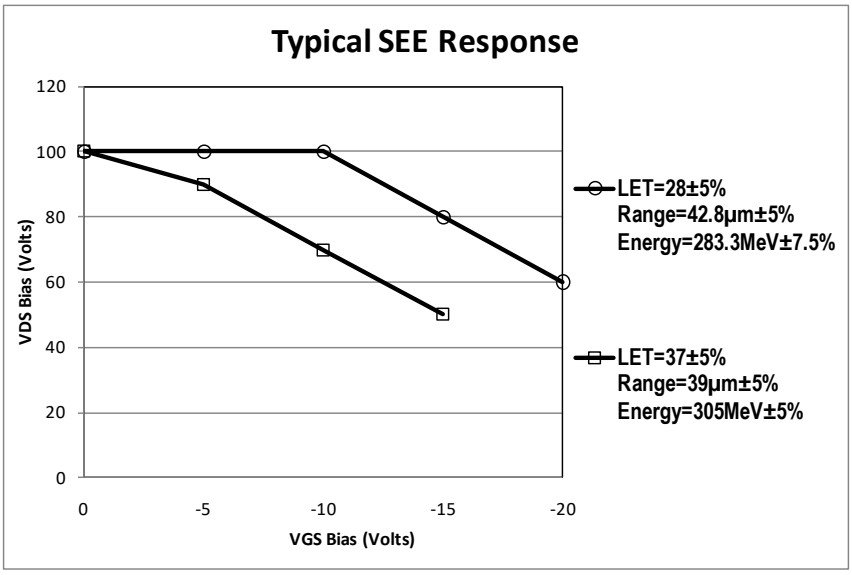
| Manufacture s cage | Inspection | MIL-STD-750 | | Sample plan |
|---|---|---|---|---|
| | | Method | Conditions | |
| 69210 (Applicable to devices with a date code of 16 June 1998 and older) | SEE 1/ | 1080 | See MIL-STD-750E method 1080.0 dated 20 November 2006. See figure 6 | |
| | Electrical measurements | | $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ in accordance with table I, subgroup 2 | 3 devices |
| | SEE irradiation: | | Fluence = 3E5 ±20 percent ions/cm$^2$<br>Flux = 2E3 to 2E4 ions/cm$^2$/sec, temperature = 25º ±5 °C | |
| * | 2N7380, 2N7380U3 | | Surface LET = 28 MeV-cm$^2$/mg ±5%, range = 42.8 µm ±7.5%, Kr ion energy = 283.3 MeV ±7.5%<br>In-situ bias conditions: $V_{DS}$ = 100 V and $V_{GS}$ = -10 V<br>$V_{DS}$ = 80 V and $V_{GS}$ = -15 V<br>$V_{DS}$ = 60 V and $V_{GS}$ = -20 V<br>(typical 4.53 MeV/nucleon at Brookhaven National Lab Accelerator) | |
| | 2N7381 | | In-situ bias conditions: $V_{DS}$ = 190 V and $V_{GS}$ = 0 V<br>$V_{DS}$ = 180 V and $V_{GS}$ = -5 V<br>$V_{DS}$ = 170 V and $V_{GS}$ = -10 V<br>$V_{DS}$ = 125 V and $V_{GS}$ = -15 V<br>(nominal 4.53 MeV/nucleon at Brookhaven National Lab Accelerator) | |
| * | 2N7380, 2N7380U3 | | Surface LET = 37 MeV-cm$^2$/mg ±5%, range = 39 µm ±5%, Cu ion energy = 305 MeV ±5%<br>In-situ bias conditions: $V_{DS}$ = 100 V and $V_{GS}$ = 0 V<br>$V_{DS}$ = 90 V and $V_{GS}$ = -5 V<br>$V_{DS}$ = 70 V and $V_{GS}$ = -10 V<br>$V_{DS}$ = 50 V and $V_{GS}$ = -15 V<br>(typical 3.77 MeV/nucleon at Brookhaven National Lab Accelerator) | |
| | 2N7381 | | In-situ bias conditions: $V_{DS}$ = 100 V and $V_{GS}$ = -10 V<br>$V_{DS}$ = 50 V and $V_{GS}$ = -15 V<br>(typical 3.77 MeV/nucleon at Brookhaven National Lab Accelerator) | |
| | Electrical measurements | | $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ in accordance with table I, subgroup 2 | |
| Upon qualification, all manufacturers should provide the verification test conditions to be added to this table. | | | | |

1/ $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ was examined before and following SEE irradiation to determine acceptability for each bias condition. Other test conditions in accordance with table I, subgroup 2, may be performed at the manufacturer's option.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

2N7380, 2N7380U3




2N7381

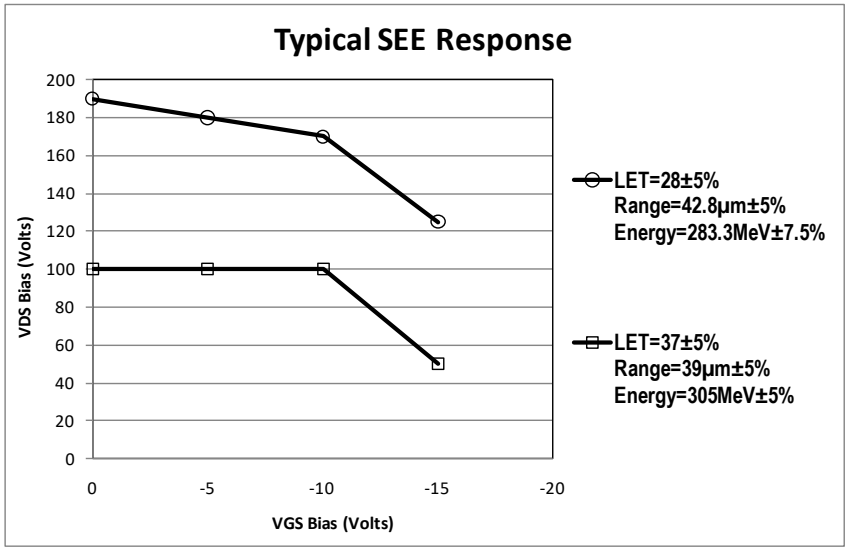



FIGURE 6. Single event effects safe operating area graphs.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.

6.9 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692-6939 or DSN 850-6939.

* 6.10 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy – SH
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - SM
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2021-092)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:09Z
Check the source to verify that this is the current version before use.